

Methodology and caveats

Cleaning and demand

Duplicate order IDs are dropped after first occurrence. Invalid dates, missing keys, non-finite quantities/prices, negative prices, unknown SKUs, and discounts outside [0, 1] are excluded. Negative quantity rows are returns; signed quantities are aggregated per SKU/day, then net daily demand is floored at zero. This assumption needs adaptation for businesses where returns are attributed to original sale dates.

Features

For each SKU, lag 1/7/14/28 and rolling means of 7/14/28 use observations strictly before the target. Calendar fields are known for the future. The model is recursively forecast: each prediction becomes history for the next date, so horizon error may compound.

Model comparison

The test set is the last test_days observations per SKU. Baselines: last-value naive, same weekday seven days ago, seven-day moving average. ML model: scikit-learn histogram gradient boosting on engineered features. Aggregate scores pool held-out SKU-days. Do not compare numbers across datasets/horizons as if they were universal. WAPE is undefined if aggregate actual demand is zero.

Inventory equations

For service level p , let $z = \text{NormalDist inverse CDF}(p)$. Safety stock approximates $z * \sigma_{\text{daily}} * \sqrt{L}$, where L is fixed supplier lead time. Reorder point is $\mu_{\text{daily}} * L + \text{safety_stock}$. EOQ is $\sqrt{2 * D * S / H}$ where D is annualized mean demand, S ordering cost, and H annual holding cost per unit. MOQ is applied to positive suggested quantity. $\text{days_of_supply} = \text{stock} / \text{mean daily demand}$; zero-mean SKUs have infinite days of supply.

ABC uses cumulative demand-value proxy (units unit cost): A through 80%, B through 95%, and C remainder. XYZ uses coefficient of variation: X 0.5, Y 1.0, Z > 1.0; thresholds are illustrative. Revenue proxy is not gross margin or business criticality.

Known limitations

No confidence intervals, multiple backtest folds, dynamic covariates, censoring correction for stockouts, open-order receipts, variable lead-time distribution, supplier calendars, pack-size constraints, budget/capacity optimization, live data, or automatic purchase execution. Current-stock values are generated assumptions. Every recommendation must be verified by a human.